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Integrated Policy Strategies for China's Aging Population and Low Fertility Challenges

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Abstract: China's unprecedented demographic transition, marked by declining fertility rates, population aging, and negative growth, poses systemic risks to its economic sustainability and social stability. This paper examines the multidimensional linkages between demographic shifts and economic systems. By analyzing China's labor market contraction, pension deficits, healthcare financing pressures, and intergenerational inequality, the study reveals how these challenges are interconnected and exacerbated by institutional inertia. Drawing comparative insights from other countries, the paper proposes a holistic policy framework including labor market reforms to promote flexible employment, pension system restructures, and family support ecosystems featuring subsidized childcare and gender-inclusive parental leave. Additionally, healthcare financing innovations such as preventive care integration are urged to address aging-related costs. The findings emphasize the urgency of transitioning from a 'demographic dividend' reliance to a quality-driven development model, ensuring equitable resource allocation across generations and regions. This research contributes actionable strategies for policymakers to navigate China's dual challenges of low fertility and rapid aging while fostering sustainable population governance.

Keywords: economic development, aging population, labor market, healthcare

1. Introduction

In recent years, demographic transformations in China have attracted significant scholarly attention due to their unprecedented nature and societal implications [1]. Official data released in early 2023 revealed that the nation had entered a phase of negative population growth for the first time in six decades. According to the China Statistical Yearbook, projections for 2024 indicate not only decelerated population expansion but also a sustained decline in total fertility rate (TFR) falling beyond expectation [2]. These shifts pose profound challenges to economic sustainability, social welfare systems, and intergenerational equity. These trends threaten to erode the 'demographic dividend' that fueled China's economic growth, while amplifying pressures on pension systems, healthcare financing, and labor markets. How to deal with low fertility rates and how to actively coping with population aging have become the focus of research.

Academic research has shifted from focusing on population quantity to population quality and structure, hoping to give full play to the advantages of human resources, transform the demographic dividend into talents dividend, and promote high-quality population development. However, current literature has predominantly focused on either quantitative demographic projections or isolated policy interventions, such as demographic response or pension reforms. Few studies holistically analyze

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how structural demographic changes interact with economic systems, nor do they adequately incorporate comparative insights from countries like Japan and South Korea, which faced similar transitions previously.

This paper addresses these gaps by examining the multidimensional linkages between demographic structure and economic development in China, and integrating international insights, in order to advocate systemic reforms on labor flexibility, pension restructuring, subsidized childcare, and preventive healthcare to transition from demographic dividend reliance to human capital-driven resilience.

2. Overview of China's demographic transition

China's demographic landscape is undergoing unprecedented transformations characterized by rapid population aging, persistently low fertility rates, and negative population growth. By the end of 2021, China's elderly population (65 years and above) has reached 200 million and has increased steadily in recent years, showing not only a large scale of aging population, but also a much faster pace of aging population growth than other aging countries [2]. According to projections from the United Nations World Population Prospects, China's population aging trajectory is expected to intensify substantially, with the proportion of elderly citizens forecasted to reach 26.1% by 2050 and rise to 30.1% by 2075. Notably, China demonstrates the most rapid demographic transition pace among industrialized nations, achieving deeply aged society status that typically spans multiple generations in comparable developed economies [3]. The phenomenon of aging before being rich is also serious. The Gross Domestic Product (GDP) per capita of China was just above $1000 when China entered an aging society. However, developed countries like Japan, had a GDP per capita of about $5,000 in 2000, indicating that they had sufficient financial resources to deal with population aging, while China was unable to afford the aging issue at a low level of guaranteed income.

To alleviate the low fertility rate, China introduced a two-child policy as early as 2015 and a three-child policy in June 2021. However, the abolition of the one-child policy may not necessarily lead to a significant rise in the birth rate. The Seventh National Population Census data indicate that China's TFR has declined to 1.3, significantly below the population replacement threshold of 2.1, and has continued to decline in subsequent years, initiating a phase of natural negative population growth. The number of women of productive age (15 to 49 years) is about 46.78 million, taking up 21.98% of the total population, reflecting the low fertility potential of China.

China's natural growth rate (NGR) fluctuated between 7.5‰ and 4.5‰ for the past two decades. However, the NGR experienced a dramatic decrease in recent years. In early 2023, the government announced that national population growth had been thrown into reverse for the first time since 1961, with an NGR of -0.60‰, from which it dropped to -1.48‰ at the end of the year. The proportion of population aged 0 to14 reached its lowest point at 16.3%, illustrating a same negative potential of China's workforce and the passing of demographic bonus [2].

3. Economic implications of demographic shifts

The demographic shifts in China, characterized by an aging population, low fertility rates, and shrinking workforce, have profound economic implications. These changes are reshaping the labor market, increasing the burden on social security systems, and exacerbating intergenerational inequality.

3.1. Labor market contraction

One of the most immediate and far-reaching economic impacts of China's demographic shift is the contraction of the labor market. Since 2015, the working-age population (ages 16 to 59) has declined

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by an average of 6.5 million annually [2]. This trend is particularly concerning for China's manufacturing sector, which has long relied on demographic bonus to continuously bring a surplus of young, low-cost labor to drive competitiveness, and export growth. The shrinking workforce not only increases labor costs but also reduces the availability of skilled workers, potentially undermining China's position in global supply chains.

Moreover, the reduction in the working-age population is accompanied by a higher-than-expected rise in dependency ratios. As shown in Figure 1, the elderly dependency ratio, which already surpassed 20% in 2021, rose to 22.5% in 2023. The gross dependency ratio reached 46.5% in the same year, which turns out that the average two-earner family in China has four elderly people and more than one child to take care of. This shift places additional pressure on the workforce to support both younger and older dependents, further straining economic productivity.

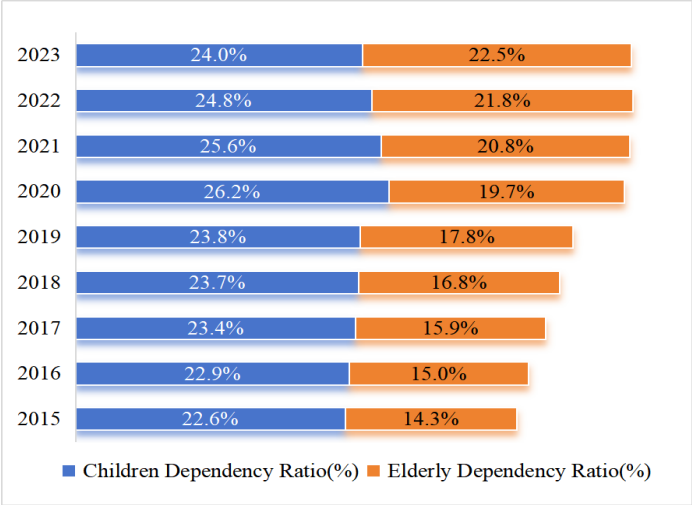

Figure 1: Dependency ratio (data from: national bureau of statistics of China)

3.2. Pressure on pension and healthcare systems

The Chinese government faces a looming crisis in funding pensions for retirees, as the shrinking workforce reduces the contributions to social security coffers while the number of beneficiaries increases.

The rapid aging of China's population has placed unprecedented strain on its pension systems, exacerbating fiscal sustainability and equity concerns. First, China's pension assets remain small, accounting for merely 9.1% of GDP—far below international benchmarks and has a narrow coverage that only 67% of urban workers and 40% of rural residents are covered [4]. Second, China's pension system faces a massive implicit debt due to legacy costs from the pre-reform era. The transition from a pay-as-you-go system to a mixed 'social pooling and personal accounts' model left unfunded liabilities. Worse, personal accounts are functionally 'empty' as contributions are diverted to finance current retirees' benefits. Third, the degree of regional inequality is relatively serious, as many rural elderly currently surviving on basic pensions averaging at 55 yuan per month under the New Rural Social Pension Scheme—far below urban levels. The pooling efficiency of pension funds is low. Affluent provinces accumulate surpluses, while aging industrial regions struggle with deficits.

The healthcare system is particularly vulnerable, as the demographic shift intensifies demand for healthcare services. Older populations require more frequent and costly medical care, with per capita health expenditures for those over 65 being 7.25 times higher than for those under 25. Concurrently, healthcare financing faces strain as the Basic Medical Insurance (BMI) fund's revenue growth struggles to match escalating expenditures. Projections indicate potential deficits by 2026, driven by

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rising retiree-to-worker ratios and systemic aging [3]. Regional disparities further compound pressures, with urban health expenditures tripling rural levels, reflecting inequitable access to resources [5]. Due to the rapid aging, chronic diseases, long-term care needs, and rehabilitation services are becoming significant challenges for China's healthcare infrastructure. Without reforms to enhance the sustainability of these systems, the financial burden on the state and individual families will continue to rise.

3.3. Intergenerational inequality

The demographic changes in China are also exacerbating intergenerational inequality, particularly between younger and older cohorts. According to the Gini coefficient released by the National Bureau of Statistics, although the income gap between Chinese residents has narrowed, it still exceeds the international Gini coefficient warning line of 0.4. The unemployment rate for workers aged 16 to 24 in urban areas reached 21.3% in 2023, while that for those aged over 24 was 4.1% [3]. It reflects the difficulties faced by young people in securing stable, well-paying jobs. This high unemployment rate is partly due to the structural mismatch between the skills of younger workers and the demands of a rapidly evolving economy.

Moreover, the rising dependency ratio and the strain on public services disproportionately affect younger generations. Young workers are forced to bear a larger share of the financial burden for elderly care and social security, limiting their ability to invest in education, housing, and other forms of personal development. This generational imbalance risks perpetuating cycles of inequality, as younger cohorts face greater economic uncertainty while older generations benefit from existing social contracts.

4. Policy responses: successes and limitations

In response to the low fertility rate, China introduced the two-child policy as early as 2015 and the three-child policy in June 2021 to incentivize childbearing. Accompanying this policy, supportive measures were rolled out, including extended maternity leave, paternity leave expansions, and tax deductions for families with multiple children. In urban areas, the policy has slightly elevated second-child birth intentions among middle-class families, particularly those benefiting from extended parental leave and workplace flexibility. Despite these efforts, these policies fail to address structural barriers that deter childbearing. A key limitation is the persistent scarcity of affordable childcare services. Only 5.5% of children under three had access to public daycare facilities, forcing families to rely on expensive private options or intergenerational care [2]. This shortage disproportionately impacts working mothers, exacerbating gender inequities. The disproportionate time and energy expenditure required by childcare responsibilities predominantly borne by women is associated with diminished earning potential and increased risks of workforce withdrawal [6]. Extended leave policies, while well-intentioned, often reinforce workplace discrimination. Additionally, urban housing prices, averaging 9.3 times annual household income in major cities, deter young couples from expanding their families [2]. These financial pressures, coupled with cultural shifts toward individualism and career prioritization, have entrenched low fertility norms.

Since the transition from a 'pay-as-you-go' system to a multi-pillar framework in 1990s, China's pension system has expanded coverage to over 1.07 billion participants by 2024 [7]. The integration of urban and rural resident pensions in 2014 mandating non-agricultural workers to join urban pension schemes bridges coverage gaps, increasing rural participation from 40% to 73.3% by 2018 [5]. Pension fund balances grew steadily, reaching 78.17 trillion yuan by 2023, supported by state-owned enterprise profit transfers. However, the pension reforms rely heavily on state-owned employees transfers and delayed retirement, yet fail to address rural poverty or incentivize private savings. The

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multi-pillar system remains underdeveloped and efforts to diversify through enterprise annuities and commercial pensions have seen limited success. Enterprise annuities cover only 23.4% of urban workers, while commercial pensions lack tax incentives to increase savings.

The healthcare financing model, centered on BMI, has achieved a universal coverage of 97.1% of the population. Medicare fund balances grew at 20% annually, reaching 2.97 trillion yuan in 2019. BMI integration in 2016 unified urban and rural schemes, reducing out-of-pocket payments from 35.29% to 28.36% of total health expenditures to effectively improve the quality of medical security [5]. Despite coverage gains, the rural elderly face unmet healthcare needs [8]. Urban health expenditures tripled rural levels, reflecting unequal access to services. Affluent provinces like Guangdong maintain surpluses, while poorer regions like Gansu face deficits. Revenue-expenditure mismatch become invisible risks. BMI for urban residents faces a retiree-to-worker ratio of 35.92%, straining fund revenues as retirees cease contribution [3]. Projections suggest deficits by 2026.

China's recent population measures reflect a recognition of the demographic crisis but fall short of addressing its root causes. Systemic limitations underscore the urgency of deeper reforms. Without tackling structural inequalities and cultural barriers, policies risk perpetuating the 'low fertility trap' and 'aging before getting rich.' Future strategies must prioritize holistic, long-term solutions over incremental adjustments to avert demographic stagnation.

5. Implications from overseas experience and recommendations

China's demographic challenges demand a holistic, adaptive policy framework. By synthesizing lessons from East Asian peers, such as enhancing labor market flexibility, optimizing social security efficiency, and cultivating inclusive family ecosystems, China can navigate its dual demographic transition.

5.1. Labor market innovation

South Korea's experience highlights the critical need to address structural labor market barriers that deter family formation. Despite multi-faceted policies, its failure to stabilize youth employment or reduce housing costs has exacerbated low fertility rates. China must prioritize labor market reforms to enhance job security and affordability for young adults, such as expanding public housing programs and enforcing wage protections for non-regular workers. Also, adopting insights from Japan's 'lifelong active society' framework presents a strategic approach to addressing population aging through enhanced labor force participation among older adults. Enabling seniors possessing the capacity and willingness to engage in sustained workforce participation beyond conventional retirement thresholds could effectively mitigate the per capita societal burden [9]. China should integrate age-friendly workplace policies, such as phased retirement and lifelong skill development programs to promote flexible work arrangements, and align seniority-based wage systems with productivity, mitigating labor shortages while leveraging older workers' skills.

5.2. Social security overhaul

A formidable long-term challenge for China lies in maintaining sustainable pension fund balances, which remain fundamental to 'a sense of security' for elderly populations as national demographic structures transition toward advanced aging stages [10]. To achieve universal security, coverage must expand through mandatory employer compliance for private-sector workers and integrate migrant workers into urban pension schemes or introduce non-contributory pensions for landless elderly, aiming to bridge the urban-rural gap. Simultaneously, China can diversify the multi-pillar system by enhancing enterprise annuities and incentivizing tax-deferred commercial pensions through simplified procedures and public awareness campaigns. Also, China should resolve implicit debt by

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separating social pooling from individual accounts and injecting state-owned asset transfers to fill funding gaps, ensuring that personal accounts are fully funded rather than diverted for current payouts.

China can establish community care networks through non-governmental organizations to provide healthcare and social services, reducing reliance on fragmented family support, while addressing poverty and healthcare gaps. Japan's aging-driven healthcare expenditure growth highlights the need for sustainable financing models. China should transition toward preventive care and digital health integration, leveraging wearable technologies to reduce hospitalization costs [11]. Reforming insurance reimbursement structures to prioritize outpatient and telemedicine services would enhance efficiency. Simultaneously, expanding long-term care insurance pilots and incentivizing private-sector investments in elderly care infrastructure could alleviate public fiscal pressures. South Korea's misallocation of fertility policy budgets warns against over-reliance on universal subsidies. Instead, China should adopt targeted financing mechanisms, such as means-tested healthcare vouchers for low-income families and public-private partnerships for elderly assistive technologies.

5.3. Family support ecosystem

Although policies aimed at stimulating the increase of fertility rate have short-term effects, they cannot fundamentally reverse the problem of low birth rate. The basic orientation of social policy is to relieve everyone from highly competitive pressure, restore the simple life normal, and have children at ease [12]. Therefore, shifting from short-term pronatalist incentives to systemic labor reforms is critical. Japan's work-life balance initiatives, including extended paternal leave to 4 weeks in 2022 and mandatory corporate transparency on leave utilization raised male leave uptake from 6% to 13%. Expanding public care infrastructure reduced unpaid domestic burdens. Japan increased community-based childcare centers from 2,737 to 6,911 and universalized free preschool for under 5-year-olds. Meanwhile, marriage subsidies and fertility medical coverage aim to counter 'no children' trend by legitimizing delayed family formation [12]. China should learn from Japan's fertility policies to build a comprehensive ecosystem featuring state-subsidized childcare centers, extended paternal leave quotas, progressive tax deductions for multi-child families, preemptively expand access to affordable fertility treatments and maternal healthcare. Furthermore, public campaigns to destigmatize non-traditional family structures and incentivize corporate adoption of family-friendly practices such as telework or on-site daycare could reduce work-family conflicts.

6. Conclusion

This study underscores the critical imperative for China to address its unprecedented demographic transition, marked by declining fertility rates and rapid population aging, which collectively threaten to destabilize economic growth, social welfare systems, and intergenerational equity. By analyzing the interplay between demographic shifts and economic systems, the paper reveals that labor market contraction, pension deficits, healthcare financing pressures, and widening intergenerational inequality are not isolated challenges but interconnected phenomena requiring systemic solutions. The analysis demonstrates that China's traditional reliance on its demographic dividend is unsustainable under current trends, necessitating a paradigm shift toward leveraging human capital quality and structural policy innovation. Drawing on comparative insights from Japan and South Korea, the study advocates for a holistic policy framework that integrates labor market flexibility, social security efficiency, family-centric support ecosystems, and healthcare financing reforms. Key recommendations include abolishing age-discriminatory employment practices, rebalancing pension schemes to incentivize delayed retirement, expanding affordable childcare infrastructure, and adopting preventive healthcare technologies. These measures aim not only to mitigate immediate

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pressures but also to foster a resilient, multigenerational society capable of sustaining long-term development.

Despite its contributions, this study has limitations inherent to its scope and methodology. Regional disparities in demographic and economic conditions within China, such as urban-rural gaps and provincial resource imbalance are insufficiently explored, potentially oversimplifying policy applicability. The extrapolation of international experiences assumes cultural and institutional comparability, which may overlook contextual nuances in China's governance model. Future research should prioritize longitudinal evaluations of policy pilot programs, particularly in underdeveloped regions, to assess scalability and equity impacts. Additionally, interdisciplinary studies integrating technological advancements such as artificial intelligence-driven labor market adaptations or digital healthcare innovations could refine policy tools. Addressing these gaps will be vital to advancing evidence-based strategies in China's quest for sustainable population governance.

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